

HW 1.3b

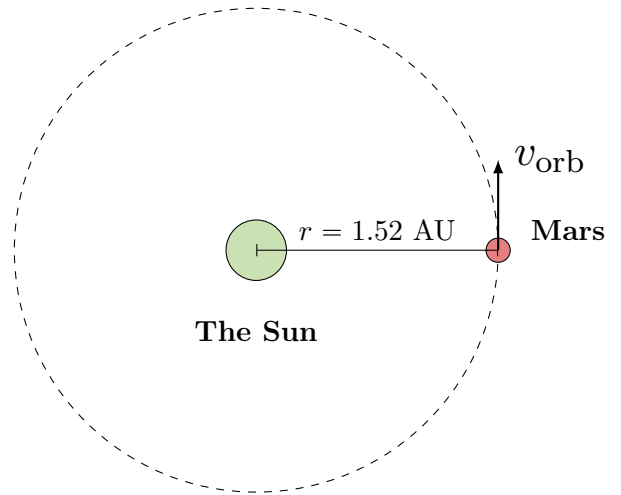
Name(s): _____

1. Mars orbits at a distance of 1.52 AU from the Sun. The length of Mars's year is 1.88 Earth years. Find the orbital velocity of Mars in :

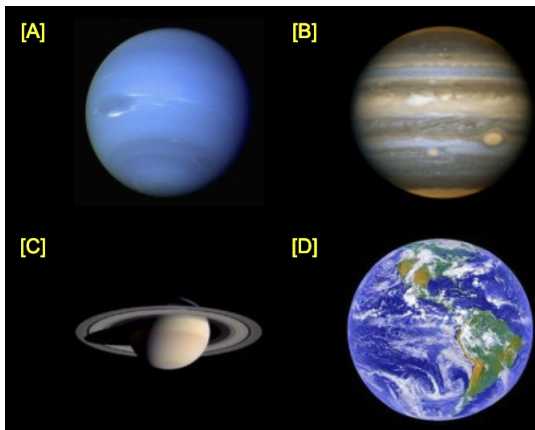
A. AU/day

B. AU/second

C. km/second

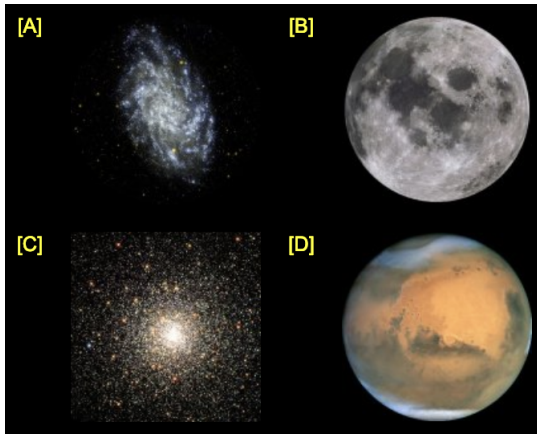


2. Which of these objects takes the longest time to orbit once around the Sun?



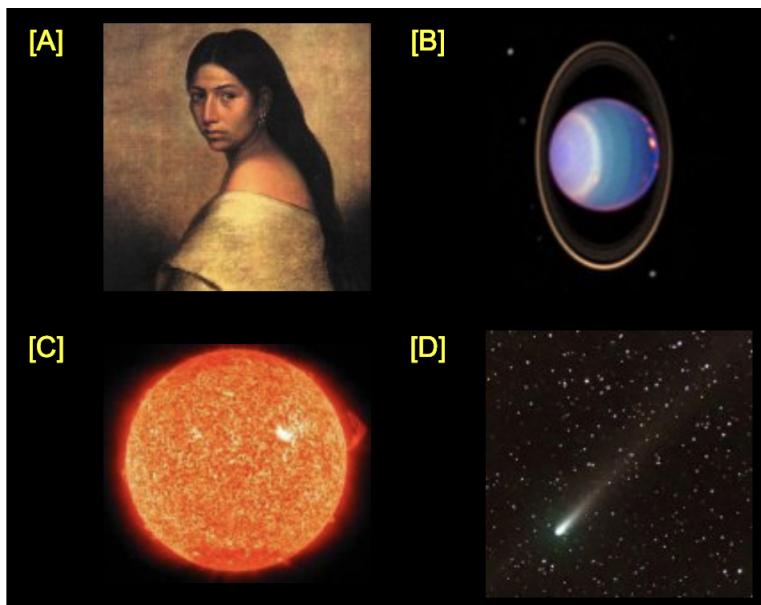
3. Approximately how many times faster is the speed of light than the orbital speed of the Earth?

4. From which of these objects does light take the shortest time to reach Earth?



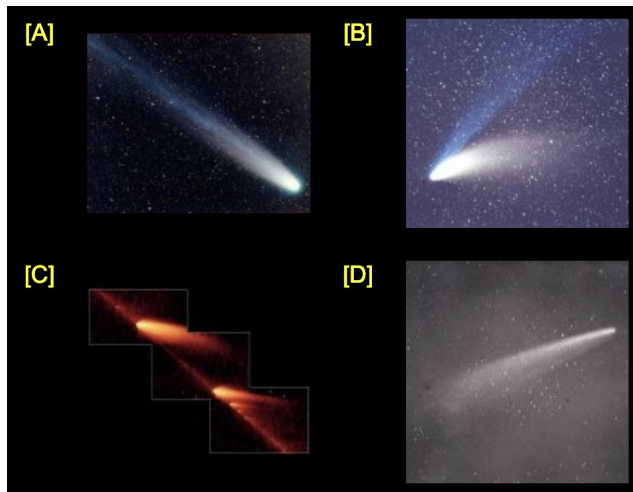
5. If the radius of the Earth is 6,400 kilometers and the radius of the Moon is 1,700 kilometers, which of the following sets of objects would make good scale models for these two bodies?
- A. soccer ball, raquetball ball C. volleyball, popcorn kernel
B. basketball, soccer ball D. orange, apple
6. If a light-year is the distance that light travels in a year at light speed, then by analogy a sound-minute is approximately how many meters? *Light travels at a speed of 3×10^5 kilometers per second, while sounds travels at roughly 300 meters per second.*
7. The radius of the Sun is 696,000 kilometers. Approximately how long does it take a photon to traverse past the Sun?
8. Roughly how long does it take the Moon to exhibit a complete set of lunar phases (its *synodic month*), in units of days?

9. How long does it take the Earth to spin once in a full circle on its own axis, in units of hours?
10. We only observe astronomical events that have happened in the past, because _____.
A. the Universe is extremely old C. that way we know where to find them
B. the speed of light is not infinite D. data reduction/analysis takes a long time
11. Which of these lasts for the least amount of time?



12. The length of the Sun's lifetime is approximately _____years.
13. The Universe has existed for _____years.

14. What type of object is shown here?



15. The Sun is located approximately 26,000 ly from the center of the Milky Way galaxy. It takes the Sun 240 Million years to orbit once around the center.

A. How far is the Sun from the Center of the Milky Way in AU?

B. How far is the Sun from the Center of the Milky Way in km?

C. How long is the path of the Sun around the Center of the Milky way in km?

D. How many seconds are in 240 Million years?

E. What is the Sun's orbital velocity around the Milky Way, in km/s?

F. How many times faster is the Sun orbiting the Milky Way than the Earth is orbiting the Sun?